

PAUL LONDON

Bioinformatics Scientist | palondon@hotmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

Bioinformatics and data science professional with 5+ years of molecular laboratory experience and 1 year of applied data science. Skilled in Next-Generation Sequencing (NGS) analysis, variant calling pipelines, and reproducible computational workflows. Proficient in Python, machine learning, and bioinformatics tools, with proven success improving lab efficiency, ensuring regulatory compliance, and building scalable pipelines. Seeking to bridge wet-lab expertise with computational approaches to advance genomic and biomedical research.

TECHNICAL SKILLS

- **Bioinformatics & Genomics:** NGS analysis, GATK, BWA, SAMtools, Picard, BEDtools, variant calling
- **Programming & Data Science:** Python (pandas, NumPy, matplotlib, seaborn, scikit-learn, XGBoost, LightGBM), statistical modeling, machine learning (ML)
- **Workflow & Platforms:** Linux, Snakemake, Git/GitHub, command-line tools, workflow automation (scripts, Excel)
- **Molecular Lab Expertise:** Molecular diagnostics (PCR, NGS), automation/robotics, assay validation, QC improvement, regulatory compliance, laboratory management

TECH PROJECTS

NGS Variant Analysis Pipeline | Bioinformatics Project | 08/25 | [GitHub](#)

- Built and tested a GATK-based NGS pipeline, enabling reproducible variant analysis in a local Linux/Python environment
- Used BWA, SAMtools, Picard, and HaplotypeCaller to process sequencing data; implemented best practices for base quality recalibration, alignment, and variant calling
- Established reproducible local Linux/Python environment for downstream variant analysis

Telecom Churn Forecast Analysis | TripleTen Project | 04/25 | [GitHub](#)

- Built binary classification models (Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost) to predict customer churn using demographic and contract data
- Achieved top performance with XGBoost (AUC-ROC: 0.901, Accuracy: 0.861) and generated actionable production-ready insights

Mining Optimization ML Prototype | TripleTen Project | 02/25 | [GitHub](#)

- Developed regression model to predict gold yield from ore using process parameters
- Improved baseline sMAPE from 9.73 to 9.38 using DecisionTreeRegressor, providing interpretable, low-error solution for process optimization

EXPERIENCE

Molecular Technologist | Elite Bio Reference Laboratory | 04/24 - Present | Tampa, FL

Independent reference laboratory leveraging advanced molecular technologies for genetic testing:

- Performed routine clinical NGS testing and optimized workflow, increasing sample pass rate to 97%
- Implemented Excel and script automation to streamline sample processing, halving turnaround time
- Led regulatory documentation efforts in startup lab environment; achieved 92% COLA inspection score
- Optimized genetic test analysis SOPs, reducing QC incidents by 75%
- Trained other personnel in best practices and workflow optimization to reduce errors

Laboratory Manager | SunPath (Medical Diagnostic Laboratories) | 01/20 - 04/24 | Tampa, FL

Independent reference laboratory utilizing heavy automation and robotics for infectious disease testing:

- Oversaw automated molecular diagnostics lab; ensured inspection readiness for CAP inspections with zero deficiencies
- Improved lab workflow efficiency by 25%, exceeding expected turnaround targets
- Supervised and trained 15 lab personnel while troubleshooting robotics and assays, maintaining high throughput and accurate testing

Client-Facing Professional Experience | 2011 – 2020

- Developed transferable skills in attention to detail, professional communication, and workflow reliability in client-facing and customer service roles

Graduate Research Assistant | University of South Florida | 01/10 - 08/11

- Conducted structural and functional studies of penicillin-binding proteins via X-ray crystallography and computational modeling

EDUCATION | CERTIFICATION

Genomic Data Science Specialization | Johns Hopkins University (Coursera) | 06/25 - Present

- Coursework: Python, Bioconductor, command-line genomic tools, DNA sequencing, variant analysis

Data Science Professional Training Program | TripleTen | 09/24 - 04/25

- Project-based training in machine learning, data visualization, and predictive modeling

Clinical Laboratory Supervisor | SU53237 (FL) – MDxT (AAB) | 2020

- Licensed by Florida; accredited by AAB/ABOR as a Molecular Diagnostics Supervisor

M.S. Biotechnology | University of South Florida | 2010 | GPA 4.0

- Focus: molecular biology, pharmaceutical manufacturing, bioinformatics applications

B.S. Biochemistry & Molecular Biology | University of Florida | 2008 | GPA 3.98 | Magna Cum Laude

- Focus: biochemistry, molecular biology, genetics, laboratory research